# Rajalakshmi Engineering College

## FUNDAMENTALS OF DATA SCIENCE LAB

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## EXPERIMENT 12

Detect outliers in a given dataset

# Experiment 12: Detect outliers in a given dataset

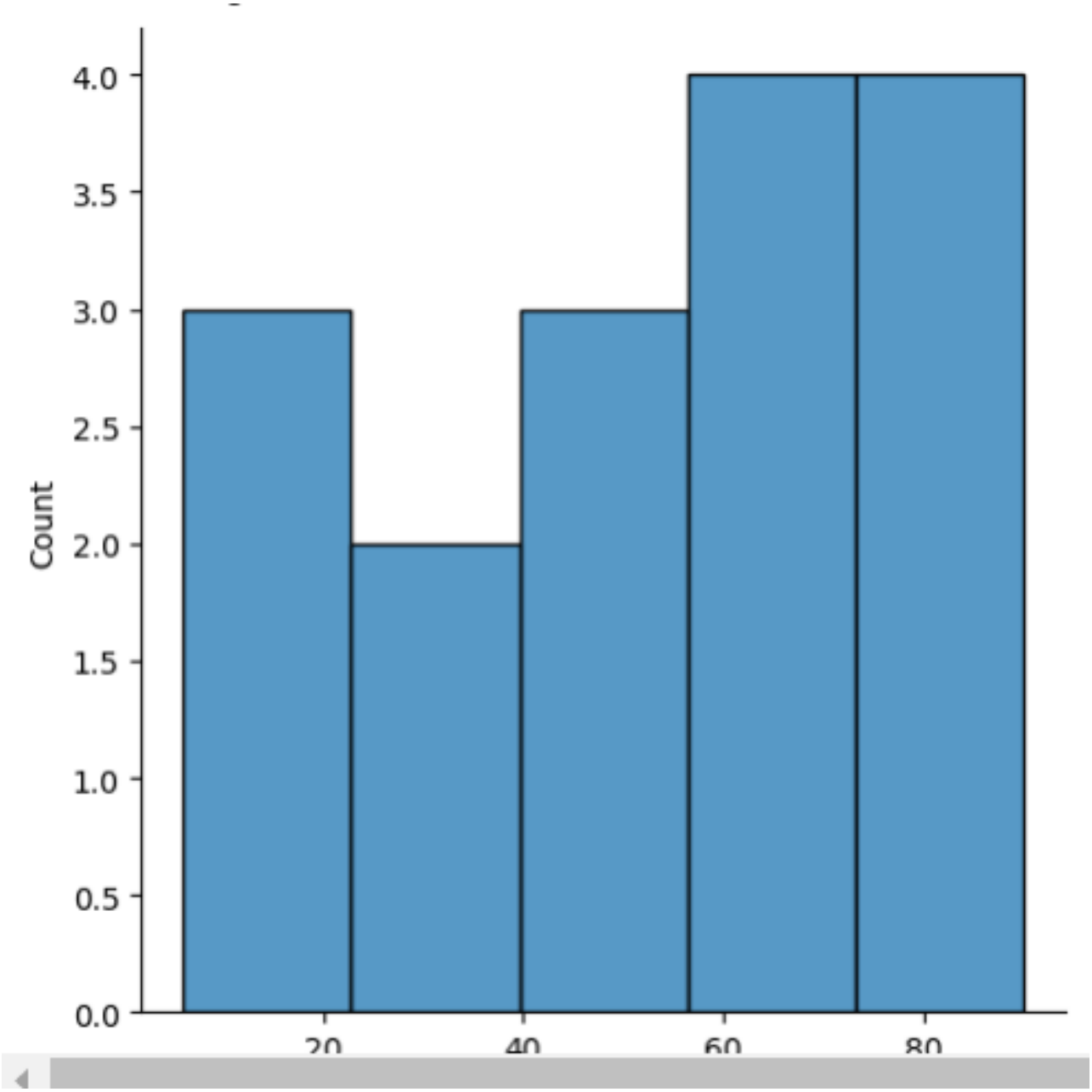
## Aim

To identify outliers in the given dataset.

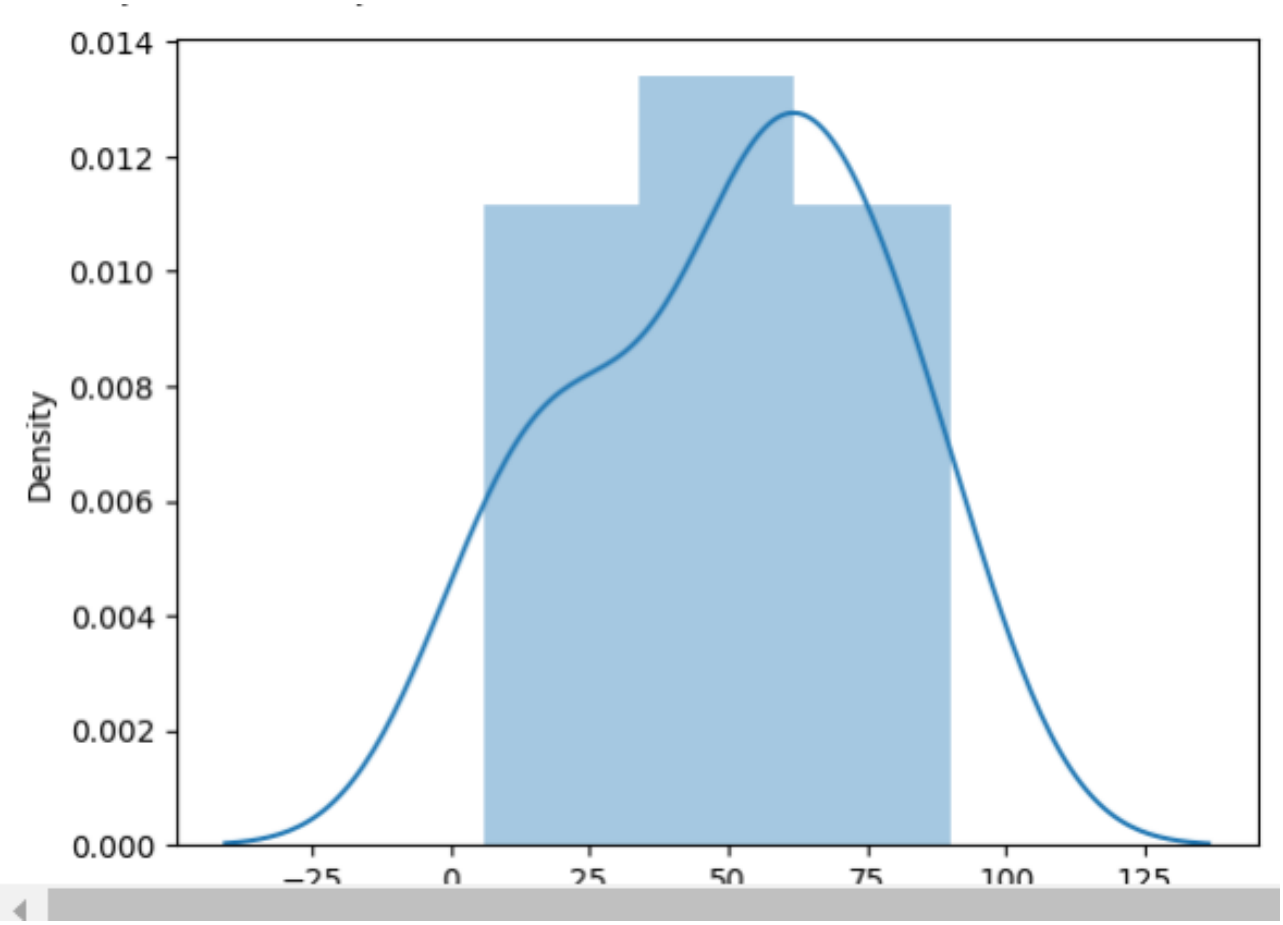
## Source Code

```
import numpy as np
array=np.random.randint(1,100,16)
array
array.mean()
np.percentile(array,25)
np.percentile(array,50)
np.percentile(array,75)
np.percentile(array,100)
def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)
import seaborn as sns
%matplotlib inline
sns.displot(array)
sns.distplot(array)
sns.distplot(array)
new_array=array[(array>lr) & (array<ur)]
new_array
sns.displot(new_array)
lr1,ur1=outDetection(new_array)
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
sns.distplot(final_array)
sns.distplot(final_array)
```

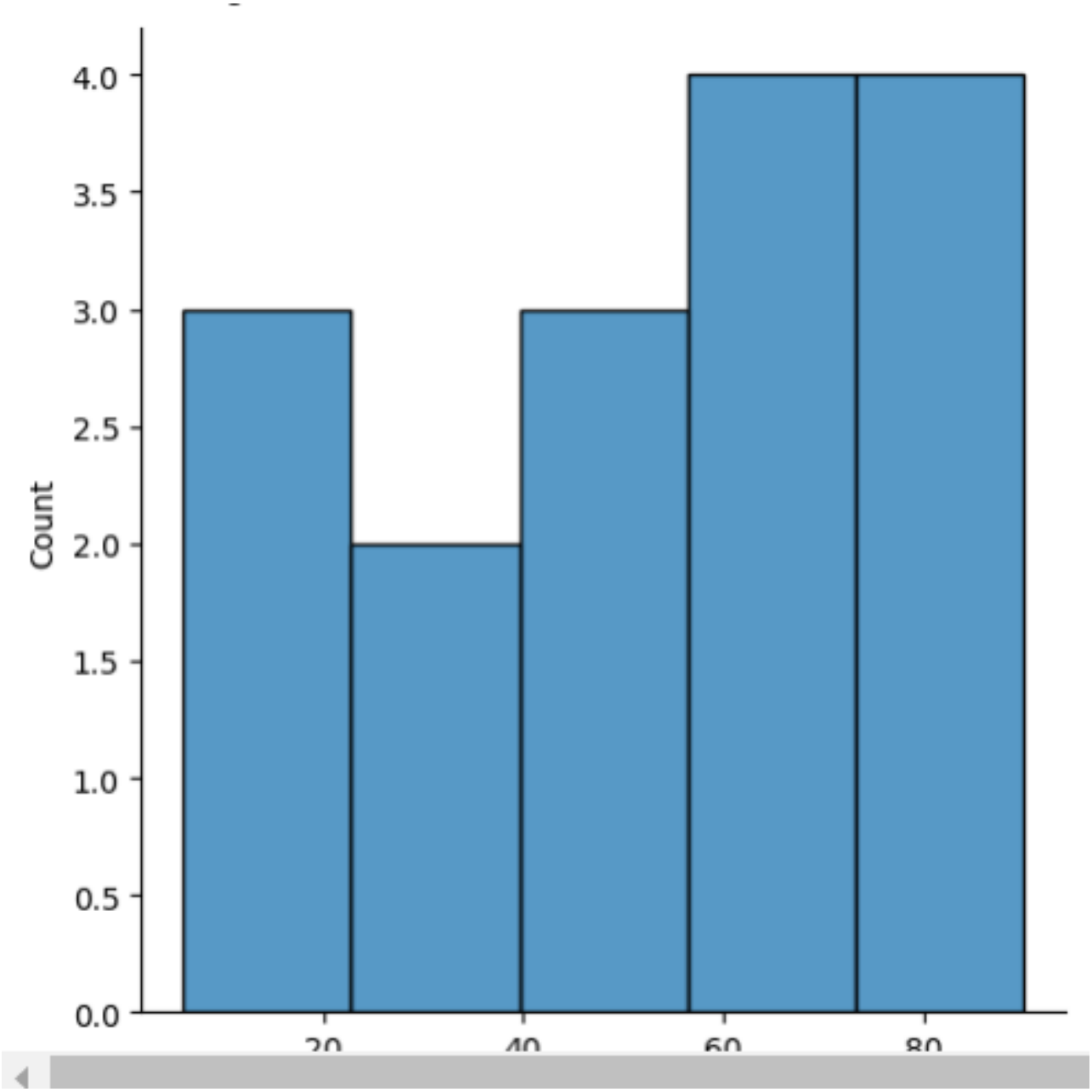
**Output - Experiment 12**



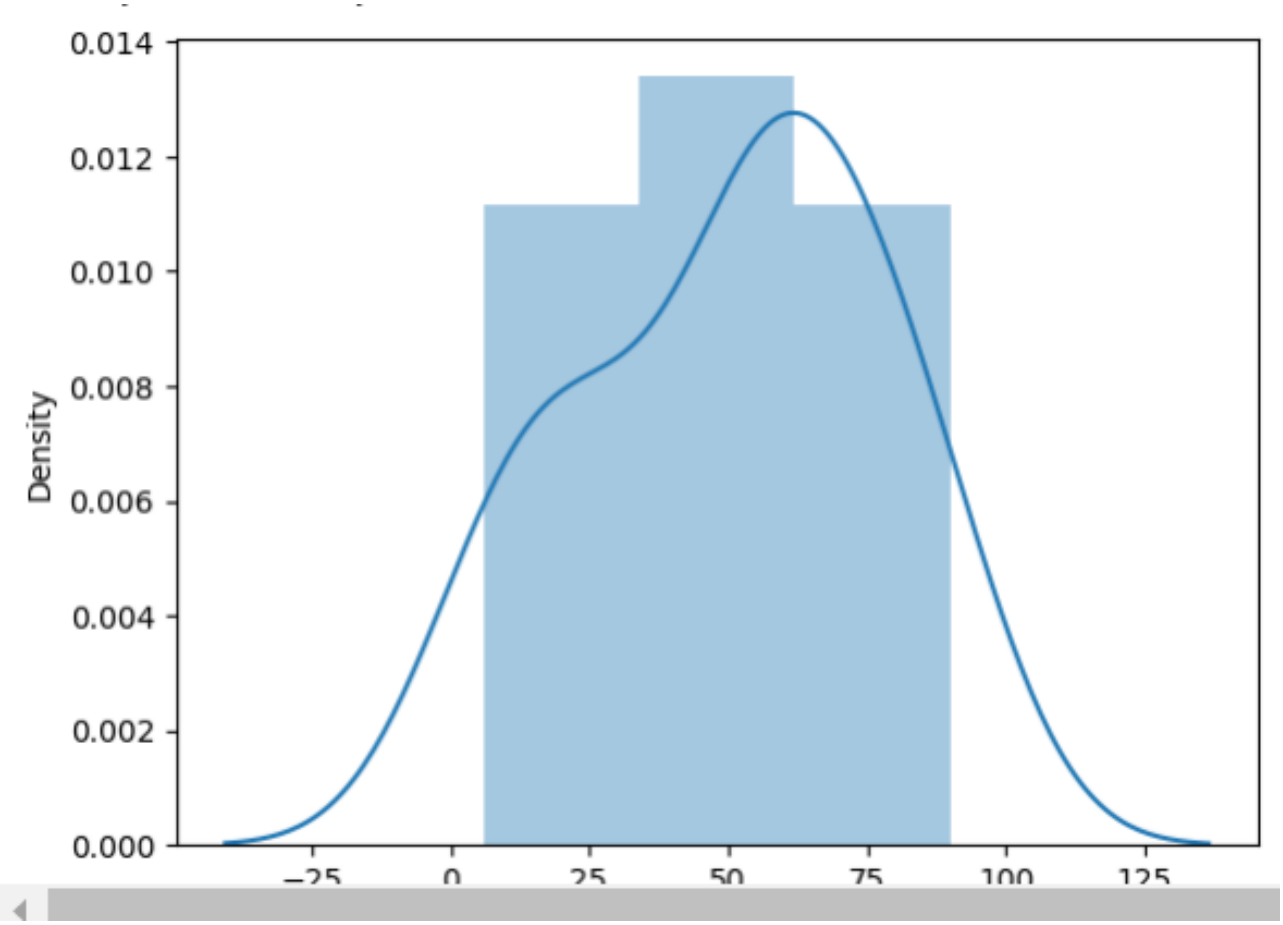
**Output - Experiment 12**



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**Output - Experiment 12**



## Result - Experiment 12

Outliers in the given dataset were detected.